Program

```
#include <stdio.h>
#define MAX 50
int graph[MAX][MAX];
int visited[MAX];
int n;
void DFS(int vertex)
{
    int i;
    // Mark vertex as visited

    visited[vertex] = 1;
    // Display vertex
    printf("%d ", vertex);
    // Visit adjacent vertices
    for (i = 0; i < n; i++)
   {
      if (graph[vertex][i] == 1 && visited[i] == 0)
      {
          DFS(i);
      }
   }
}
int main()
{
   int i, j;
   int start;
   // Read number of vertices
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   // Read adjacency matrix
   printf("Enter the adjacency matrix:\n");
   for (i = 0; i < n; i++)
   {
      for (j = 0; j < n; j++)
     {
         scanf("%d", &graph[i][j]);
      }
   }
   // Initialize visited array
   for (i = 0; i < n; i++)
   {
     visited[i] = 0;
```

```
    }
    // Read starting vertex
    printf("Enter starting vertex: ");
    scanf("%d", &start);
    printf("DFS Traversal: ");
    DFS(start);
    return 0;
}
```

Sample input

Using the same graph as the BFS example:

```
Enter number of vertices: 5
Enter the adjacency matrix:
0 1 1 1 0
1 0 1 0 0
1 1 0 0 1
1 0 0 0 0
0 0 1 0 0
Enter starting vertex: 0
```

Output

```
DFS Traversal: 0 1 2 4 3
```